

EqF for Drones

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May the Force be with you



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Purpose

- Real Time attitude and gyroscope bias estimation using IMU and magnetometer measurements.
- 3 sensors: gyroscope, accelerometer and magnetometer.
 - Magnetometer can only estimate yaw reliably.
 - Accelerometer can estimate only roll and pitch.
 - Accelerometer experiences forces other than gravity.

Filter Setup

State: $SO(3) \times so(3)$ – attitude Lie group element + lie algebra element for bias (which is angular velocity)

Symmetry Lie Group: $SO(3) \times so(3)$, basically $SE(3)$

Let $\xi = (R, b)$ – element of state space, and $X = (A, a)$ – element of Lie Group

w – angular velocity input, y_1, y_2 – measurement vectors (for acceleration and magnetic heading)

Assuming no bias random walk.

State evolution:

$$\dot{\xi} = f_w(t)(\xi) = (\dot{R}, 0) = (R(w - b)^\wedge, 0)$$

EqF functions

According to EqF paper we need to define a couple functions:

- state action
- input symmetry
- equivariant lift
- output map
- output action
- Error dynamics
- Origin ξ^o , local coordinates ϑ

State Action

$$\phi: G \times M \rightarrow M$$

$$\phi(X, \xi) = \phi((A, a), (R, b)) = (RA, A^T(b + a))$$

$$D_{\phi_X}(\xi) = (\dot{R}A, 0)$$

Input symmetry

Need to find function $\psi: G \times L \rightarrow L$ that satisfies $D_{\phi_X} f_w(\xi) = f_{\psi_X(w)}(\phi_X(\xi))$

$$D_{\phi_X} f_w(\xi) = D_{\phi_X}(R(w - b)^\wedge, 0) = (R(w - b)^\wedge A, 0) = (RAA^T(w - b)^\wedge A, 0) = (RA(A^T(w - b))^\wedge, 0)$$

$$f_{\psi_X(w)}(\phi_X(\xi)) = f_{\psi_X(u)}(RA, A^T(b + a)) = (RA(\phi_X(u) - A^T(b - a)^\wedge), 0)$$

$$(RA(\phi_X(u) - A^T(b + a)^\wedge), 0) \text{ need to match } (RA(A^T(w - b))^\wedge, 0)$$

$$\phi_X(u) = A^T(u + a)$$

Equivariant Lift

Lift $\Lambda: M \times L \rightarrow g$ satisfies $D_{\{X|id\}\phi_\xi}(X)[\Lambda(\xi, w)] = f_w(\xi)$ and $\Lambda(\phi(X, \xi), \psi(X, w)) = Ad_{X^{-1}}\Lambda(\xi, w)$

$$D_{\{X|id\}\phi_\xi}(X) = (R\dot{A}, \dot{A}^T(b + a)^\wedge + A^T\dot{a})$$

At $id\ X = (I, 0)$

$$(R\dot{A}, \dot{A}^T(b + a)^\wedge + A^T\dot{a}) = (R\dot{A}, \dot{A}^T(b)^\wedge + \dot{a}) = (R\dot{A}, -\dot{A}^T \times b + \dot{a})$$

Let $\Lambda(\xi, w) = (n, v)$, then

$$D_{\{X|id\}\phi_\xi}(X)[\Lambda(\xi, w)] = (Rn, -n^T \times b + v)$$

This should satisfy $(Rn, -n^T \times b + v) = (R(w - b)^\wedge, 0)$

$$n = (w - b)^\wedge, v = (w - b)^\wedge \times b = w \times b$$

$$\Lambda(\xi, w) = ((w - b)^\wedge, u \times b)$$

Equivariant Lift

Want: $\Lambda(\phi(X, \xi), \psi(X, w)) = \text{Ad}_{X^{-1}} \Lambda(\xi, w)$

$$\phi(X, \xi) = (RA, A^\top(b + a)), \quad \psi(X, w) = A^\top(w + a)$$

$$\text{LHS, 1st comp: } (A^\top(w + a) - A^\top(b + a))^\wedge = (A^\top(w - b))^\wedge$$

$$\begin{aligned} \text{LHS, 2nd comp: } A^\top(w + a) \times A^\top(b + a) &= A^\top((w + a) \times (b + a)) \\ &= A^\top(w \times b + w \times a - b \times a) \end{aligned}$$

$$\text{Ad}_{X^{-1}}(\mu, \nu) = (A^\top \mu, A^\top(\nu + \mu \times a))$$

RHS with $(\mu, \nu) = (w - b, w \times b)$:

$$\begin{aligned} &(A^\top(w - b), A^\top(w \times b + (w - b) \times a)) \\ &= (A^\top(w - b), A^\top(w \times b + w \times a - b \times a)) \end{aligned}$$

LHS = RHS

Output Map and Action

Want: $\rho(X, h(\xi)) = h(\phi(X, \xi))$

$$h(\xi) = \begin{pmatrix} R^\top m_{\text{ref}} \\ R^\top g_{\text{ref}} \end{pmatrix}, \quad \phi(X, \xi) = (RA, A^\top(b + a))$$

$$\text{RHS: } h(\phi(X, \xi)) = \begin{pmatrix} (RA)^\top m_{\text{ref}} \\ (RA)^\top g_{\text{ref}} \end{pmatrix} = \begin{pmatrix} A^\top R^\top m_{\text{ref}} \\ A^\top R^\top g_{\text{ref}} \end{pmatrix}$$

$$\text{Define } \rho(X, y) = \begin{pmatrix} A^\top y_1 \\ A^\top y_2 \end{pmatrix}$$

$$\text{LHS: } \rho(X, h(\xi)) = \begin{pmatrix} A^\top R^\top m_{\text{ref}} \\ A^\top R^\top g_{\text{ref}} \end{pmatrix}$$

LHS = RHS

Error State and Output coordinates

EqF error: $e = \phi_{\hat{X}^{-1}}(\xi)$.

Inverse of the state: $\hat{X}^{-1} = (\hat{A}^\top, -\hat{A}^\top \hat{a})$

So $e = \phi_{(\hat{A}^\top, -\hat{A}^\top \hat{a})}(R, b)$

$$e = \left(R\hat{A}^\top, (\hat{A}^\top)^\top (-\hat{A}^\top \hat{a} + b) \right)$$

$$e = (e_R, e_b) = \left(R\hat{A}^\top, \hat{A}b - \hat{a} \right)$$

Origin: $\xi^\circ = (I, 0)$

Now convert the error to local vector coordinates

$$\eta_R = \log(e_R)^\vee \in R^3 \quad \eta_b = e_b \in R^3$$

$$\varepsilon = \vartheta(e) = \begin{bmatrix} \eta_R \\ \eta_b \end{bmatrix} = \begin{bmatrix} \log(e_R)^\vee \\ e_b \end{bmatrix} \in R^6$$

$$\varepsilon = \begin{bmatrix} \log(R\hat{A}^\top)^\vee \\ \hat{A}b - \hat{a} \end{bmatrix}$$

Output coordinates

Inverse $\vartheta^{-1} : R^6 \rightarrow SO(3) \times R^3$

$$\vartheta^{-1}(\varepsilon) = (\exp(\eta_R^\wedge), \eta_b)$$

E_qF matrices

Now we have everything to get matrices for E_qF using these equations:

$$\mathring{A}_t := D_e|_{\mathring{\xi}} \vartheta(e) \cdot D_E|_{\text{id}} \phi_{\mathring{\xi}}(E) \cdot D_e|_{\mathring{\xi}} \Lambda(e, \mathring{u}) \cdot D_\varepsilon|_0 \vartheta^{-1}(\varepsilon). \quad (25)$$

$$C_t^\star \varepsilon = \frac{1}{2} \left(D_E|_{\text{id}} \rho(E, y) + D_E|_{\text{id}} \rho(E, \hat{y}) \right) \text{Ad}_{\hat{X}^{-1}} \varepsilon^\wedge \quad (35)$$

$$\begin{aligned} B_t &:= D_e|_{\mathring{\xi}} \vartheta(e) \cdot D_E|_{\text{id}} \phi_{\mathring{\xi}}(E) \cdot \text{Ad}_{\hat{X}} \cdot D_u|_{u_m} \Lambda(\hat{\xi}, u) \\ &= D_e|_{\mathring{\xi}} \vartheta(e) \cdot D_E|_{\text{id}} \phi_{\mathring{\xi}}(E) \cdot D_w|_{\mathring{u}_m} \Lambda(\mathring{\xi}, w) \cdot \psi_{\hat{X}^{-1}}. \end{aligned} \quad (42)$$

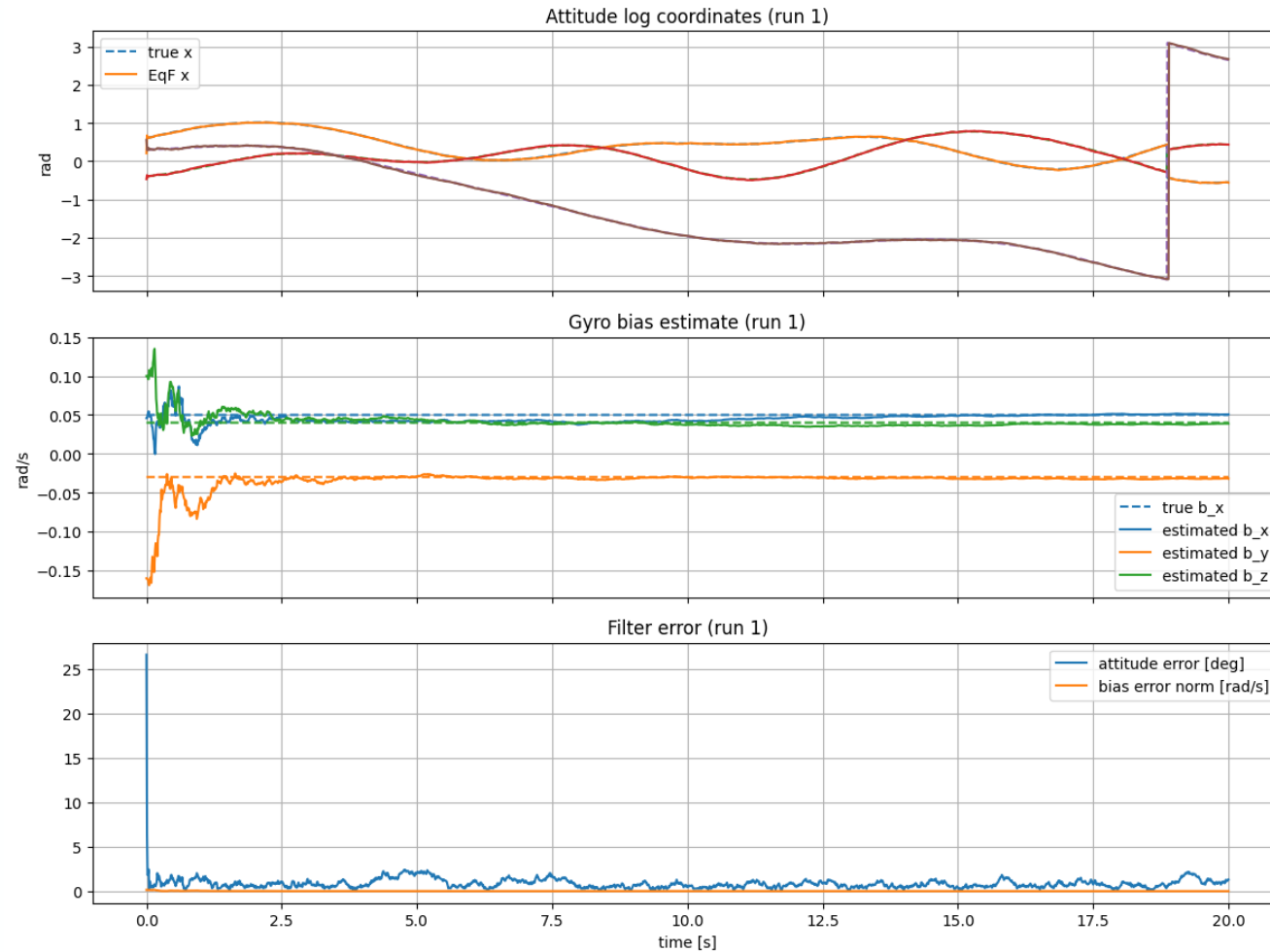
Kalman Filter Equations

$$\dot{\hat{X}} = DL_{\hat{X}}\Lambda(\phi(\hat{X}, \hat{\xi}), u) + DR_{\hat{X}}\Delta, \quad \hat{X}(0) = \text{id} \quad (36)$$

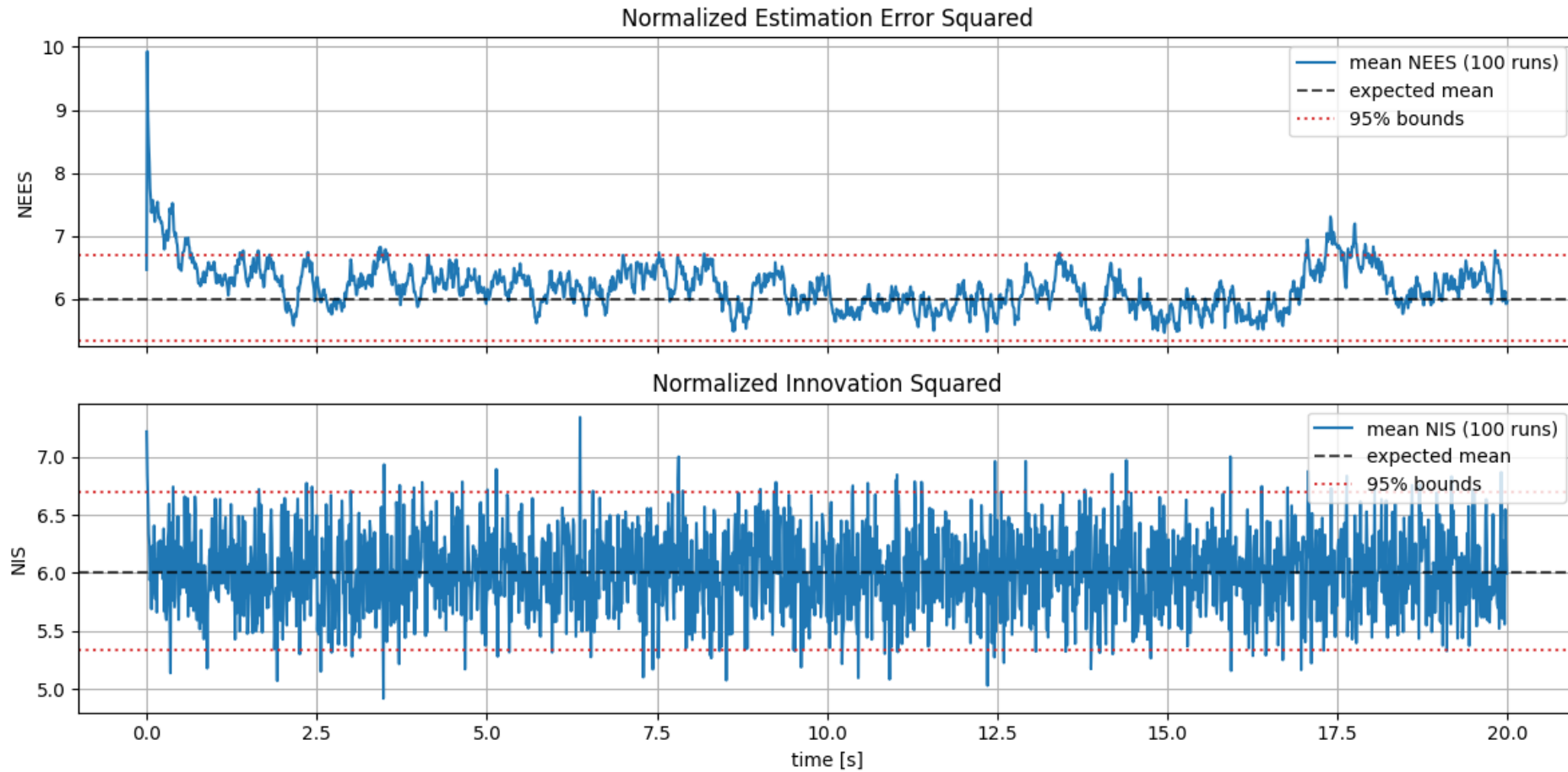
$$\Delta = D_E|_{\text{id}}\phi_{\hat{\xi}}(E)^{\dagger}D\vartheta^{-1}\Sigma C_t^{\top}N_t^{-1}(y - h(\phi(\hat{X}, \hat{\xi}))) \quad (37)$$

$$\dot{\Sigma} = \mathring{A}_t\Sigma + \Sigma\mathring{A}_t^{\top} + M_t - \Sigma C_t^{\top}N_t^{-1}C_t\Sigma, \quad \Sigma(0) = \Sigma_0. \quad (38)$$

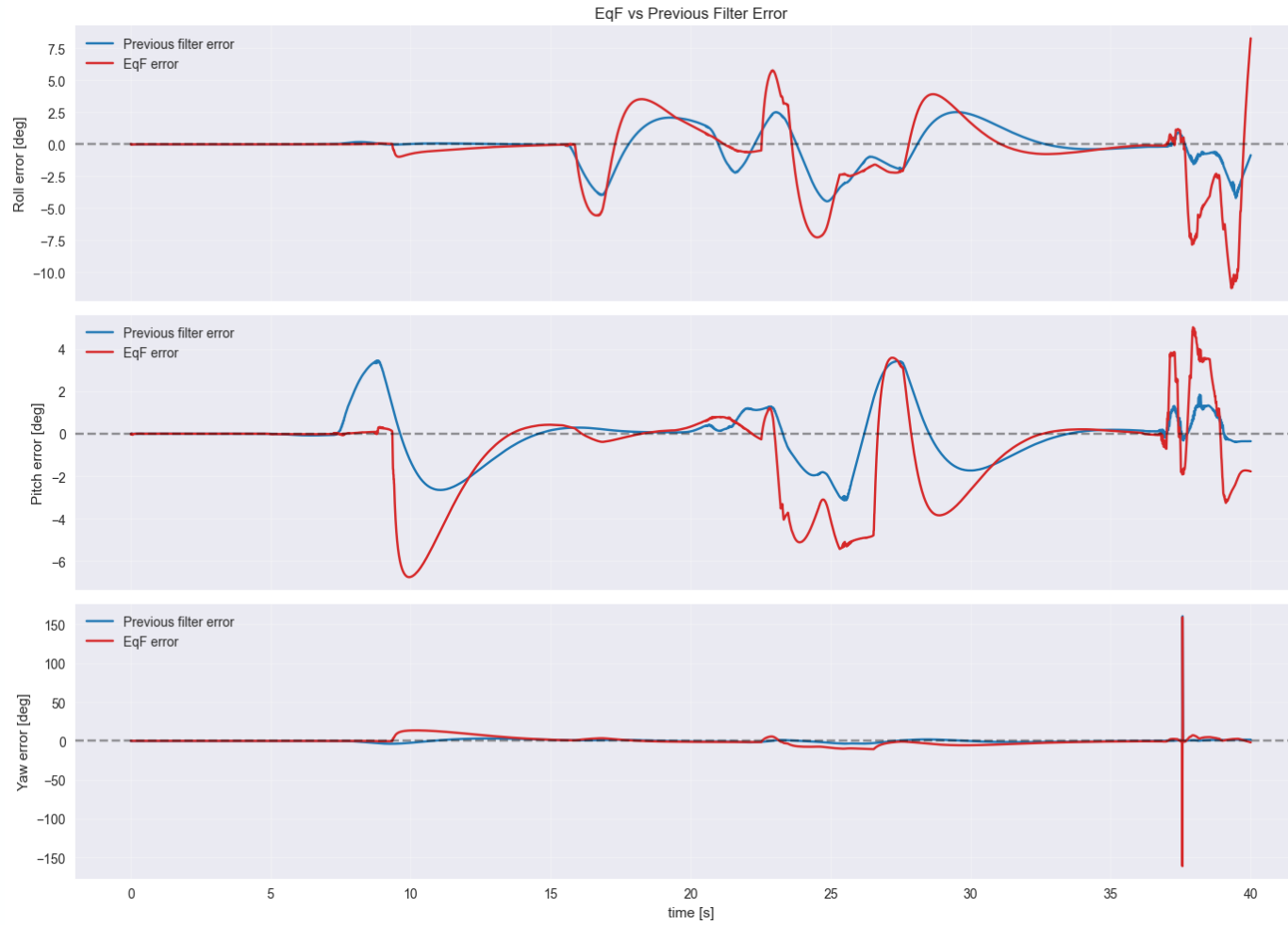
Simulation Results



NEES and NIS



Drone data



Drone Data

=== EQF VS PREVIOUS FILTER RMS ERROR ===

Roll : EqF = 2.49 deg, Previous = 1.39 deg

Pitch: EqF = 2.25 deg, Previous = 1.24 deg

Yaw : EqF = 5.58 deg, Previous = 3.42 deg

NEES and NIS for drone data

